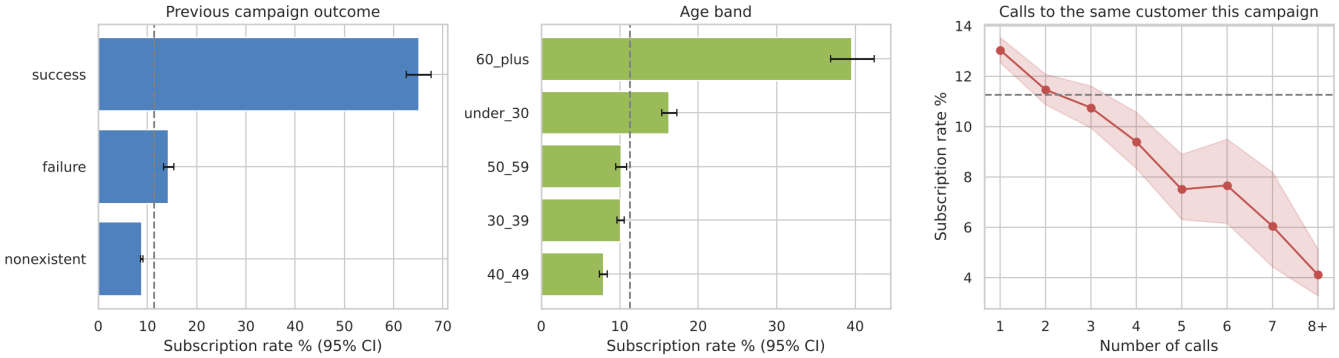
Decision Brief: Who to call in the next term-deposit campaign

Prepared by Joy Bello · 7 November 2026 · Data: UCI Bank Marketing, 41,176 cleaned call records (2008–2010) · Supporting files: notebook, cleaned dataset, data dictionary

Decision: Our telemarketing calls convert at only **11.3%**. **Recommendation:** call a defined priority segment first, stop after 3 attempts per customer, prefer mobile numbers, and confirm the gain with a 4-week controlled pilot before full rollout.

Key findings (only information known before a call is used)

Finding	Evidence	Implication
1. Past success predicts future success	Customers who said yes in a previous campaign subscribe at **65.1%** (95% CI 62.6–67.6%), 5.8× average	Call them first, every campaign
2. Some groups respond far better	Age 60+: **39.6%**; students: **31.4%**; retired: **25.3%**. Blue-collar: 6.9%; age 40–49: 7.9%	Prioritise these groups; review the message for low-response groups
3. Repeat calls have diminishing returns	Rate falls from 13.0% on 1st call to 4.1% at 8+ calls. Calls after the 3rd = **26.5% of all calls** but only **12.0% of subscriptions**	Cap at 3 attempts and use the freed capacity on priority customers
4. Mobile beats landline	Cellular **14.7%** vs telephone 5.2%	Collect and use mobile numbers
Combined effect	Priority segment (past success OR 60+ OR student/retired, on mobile) is **7.7%** of customers but delivers **29.1%** of subscriptions, converting at **42.7%** vs 8.7% for everyone else	Small list, large share of results



Subscription rate by segment, with 95% confidence intervals; dashed line = 11.3% average.

Data quality: what was fixed before analysis

12 duplicate rows removed · 12,716 hidden missing values coded *'unknown'* kept as an explicit category, not guessed (the Week 3 check had missed them) · 39,673 '999' placeholder codes in *pdays* converted to blanks · 4,110 rows where *pdays* contradicted *previous* resolved by trusting *previous* · *duration* (call length) excluded from decisions because it is only known after the call (leakage) · 12 automated validation checks passed.

Assumptions and limitations

Assumptions: duplicates are data-entry repeats (no customer ID to confirm); 'unknown' means not recorded; very high call counts are genuine records. **Limitations:** (1) one Portuguese bank in 2008–2010, so interest rates and channels differ today; (2) no year column, so crisis-era effects are mixed into months; (3) these are **associations, not causes**: past-success customers may simply be more engaged savers; (4) no cost or deposit-value data, so this optimises conversion, not profit; (5) only 11.3% positives, so small segments carry wider uncertainty.

Next step

Run a 4-week pilot: randomly split the call list into a **control** group (current practice) and a **test** group (priority-first, 3-call cap, mobile preferred). Compare subscriptions per 100 calls and per agent-hour. Add call-cost and deposit-value data so the next review can measure return, not just conversion.